

MIHIR NILESH HOLMUKHE

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Summary

Data Engineer & AI/ML Engineer with 5+ years of experience building cloud-native data platforms, distributed ETL pipelines, real-time streaming architectures, machine learning systems, and Generative AI applications across telecommunications, banking, and SaaS domains. Strong expertise in Python, SQL, Apache Spark, Kafka, Airflow, BigQuery, Snowflake, TensorFlow, LangChain, FastAPI, AWS, and GCP. Experienced in designing scalable data pipelines, Retrieval-Augmented Generation (RAG) systems, vector search, LLM applications, cloud-native analytics platforms, and production-ready AI solutions processing millions of enterprise records.

Technical Skills

Programming: Python, SQL, Java

Data Engineering: Apache Spark, PySpark, Kafka, Airflow, Apache NiFi, Hadoop, HDFS, Spark Structured Streaming, ETL/ELT, Data Modeling, Data Warehousing, Batch & Real-Time Processing

AI / Machine Learning: TensorFlow, Scikit-learn, NLP, Transformers, Deep Learning, Feature Engineering, Recommendation Systems, Predictive Analytics, Model Deployment

Generative AI: LLMs, LangChain, LangGraph, Retrieval-Augmented Generation (RAG), Prompt Engineering, OpenAI APIs, Vector Databases, pgvector, RAGAS, AI Agents

Cloud Platforms: AWS (S3, EC2, Lambda, Glue), Google Cloud Platform (BigQuery, Dataproc, Cloud Composer)

Backend & APIs: FastAPI, REST APIs, Microservices, Docker, Redis, CI/CD

Databases: BigQuery, Snowflake, PostgreSQL, Oracle, MySQL, Apache Kudu

Analytics: Tableau, Power BI, Looker, SQL Analytics, Product Analytics, Cohort Analysis, A/B Testing

Tools: Git, GitHub, Jira, Agile Scrum

Professional Experience

Software Engineer I

Jun 2025 – Present

Prodapt

Richardson, TX

- Developed scalable ETL pipelines using Apache NiFi, Spark, Kafka, and Airflow processing 12M+ telecom events daily across multi-terabyte enterprise datasets.
- Optimized Spark workloads on Google Cloud Dataproc reducing ETL runtime by 40% while improving reporting SLAs and analytics delivery.
- Built automated Airflow orchestration with DataBuck validation improving enterprise data quality and reducing manual reconciliation efforts.
- Integrated Hadoop/HDFS platforms with BigQuery and Apache Kudu enabling centralized analytics across subscriber, billing, and network datasets.
- Designed near real-time streaming pipelines using Kafka and Spark Structured Streaming reducing reporting latency and accelerating operational monitoring.
- Collaborated with analytics and platform teams to modernize Verizon's cloud-native big data architecture using Spark, BigQuery, GCP, and CI/CD pipelines.

AI/ML Intern

Jan 2025 – Jun 2025

Morgan Stanley

New York, NY

- Developed enterprise Retrieval-Augmented Generation (RAG) solutions using LangChain, OpenAI APIs, FastAPI, and pgvector reducing analyst document lookup time by 35%.
- Built agentic AI workflows with LangGraph automating document classification, compliance analysis, and multi-step reasoning across financial datasets.
- Developed FastAPI microservices supporting LLM inference, vector search, asynchronous processing, and enterprise AI applications.
- Implemented prompt evaluation, RAGAS validation, structured output testing, and hallucination reduction techniques improving LLM response quality.
- Collaborated with data engineering teams integrating Spark, Kafka, AWS, and vector databases into scalable enterprise AI platforms.

Machine Learning Engineer

Sep 2021 – Jul 2023

Space Infolab

Mumbai, India

- Engineered distributed machine learning and data pipelines using Apache Spark, Kafka, Airflow, AWS, and Snowflake processing 20M+ SaaS events monthly while reducing model retraining time by 45%.

- Developed recommendation and personalization models using Python, TensorFlow, FastAPI, and Redis improving user engagement by 27% and increasing subscription conversions.
- Built scalable feature engineering pipelines integrating Spark Structured Streaming and Snowflake enabling low-latency behavioral predictions across enterprise SaaS products.
- Implemented NLP automation workflows using Transformers, Kafka, and cloud ETL frameworks automating 70% of customer support ticket classification.
- Developed production-ready REST APIs for machine learning inference supporting scalable deployment of AI services across distributed environments.
- Partnered with engineering, analytics, and product teams following Agile methodologies to deploy scalable AI and analytics solutions.

Data Scientist

Feb 2020 – Aug 2021

Space Infolab

Mumbai, India

- Developed churn prediction and customer segmentation models using Python, SQL, and Scikit-learn on 5M+ customer records reducing subscription churn by 18%.
- Built Tableau dashboards and SQL analytics reports enabling leadership to identify low-conversion customer journeys and improve paid-user conversion by 22%.
- Performed A/B testing, statistical analysis, feature engineering, and forecasting supporting data-driven product optimization initiatives.
- Designed scalable ETL pipelines integrating multiple enterprise data sources supporting analytics, reporting, and machine learning workflows.
- Collaborated with business stakeholders translating analytical findings into actionable recommendations improving customer retention and product performance.

Projects

Enterprise GenAI Knowledge Assistant – LangChain – OpenAI – FastAPI – pgvector

- Developed enterprise Retrieval-Augmented Generation (RAG) platform integrating LangChain, OpenAI APIs, pgvector, semantic search, and prompt engineering.
- Implemented document chunking, embedding generation, vector indexing, retrieval optimization, and response validation improving answer relevance and groundedness.
- Designed evaluation pipelines using RAGAS metrics, prompt testing, hallucination detection, and response quality analysis.

Real-Time Telecom Analytics Platform – Spark – Kafka – BigQuery – Airflow

- Designed distributed Spark and Kafka streaming pipelines processing millions of telecom events supporting real-time operational analytics.
- Built automated ETL orchestration using Airflow integrating BigQuery, Oracle, Apache Kudu, and enterprise reporting platforms.
- Optimized distributed data processing reducing reporting latency while improving scalability and enterprise data quality.

Recommendation Engine – TensorFlow – FastAPI – Redis

- Developed recommendation models using TensorFlow, feature engineering, and behavioral analytics increasing user engagement by 27%.
- Built FastAPI inference services with Redis caching enabling scalable low-latency prediction APIs for enterprise SaaS applications.
- Implemented monitoring, model evaluation, and performance optimization improving recommendation quality and deployment reliability.

Education

Syracuse University

Aug 2023 – May 2025

Master of Science in Applied Data Science

Syracuse, NY

University of Mumbai

Aug 2018 – May 2022

Bachelor of Engineering in Computer Science

Mumbai, India

Certifications

- AWS Certified Machine Learning – Associate
- AWS Certified Cloud Practitioner